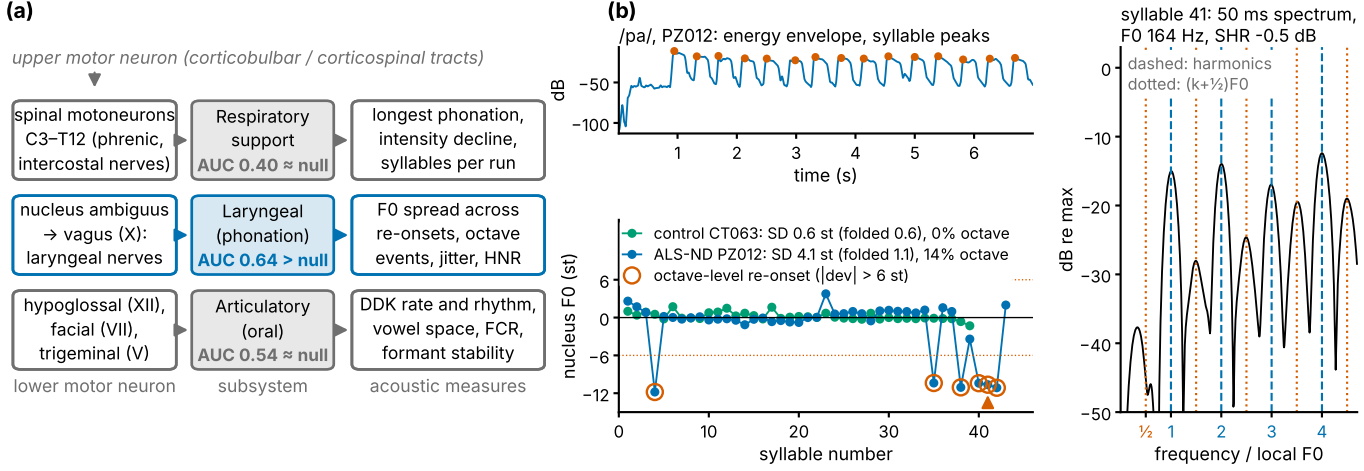
(a)
upper motor neuron (corticobulbar / corticospinal tracts)
spinal motoneurons C3–T12 (phrenic, intercostal nerves)
Respiratory support
AUC 0.40 ≈ null
longest phonation, intensity decline, syllables per run
nucleus ambiguus → vagus (X): laryngeal nerves
Laryngeal (phonation)
AUC 0.64 > null
F0 spread across re-onsets, octave events, jitter, HNR
hypoglossal (XII), facial (VII), trigeminal (V)
Articulatory (oral)
AUC 0.54 ≈ null
DDK rate and rhythm, vowel space, FCR, formant stability
lower motor neuron
subsystem
acoustic measures
(b)
/pa/, PZ012: energy envelope, syllable peaks
dB
−50
−100
1
2
3
4
5
6
time (s)
control CT063: SD 0.6 st (folded 0.6), 0% octave
ALS-ND PZ012: SD 4.1 st (folded 1.1), 14% octave
octave-level re-onset (|dev| > 6 st)
nucleus F0 (st)
6
0
−6
−12
0
10
20
30
40
syllable number
syllable 41: 50 ms spectrum,
F0 164 Hz, SHR -0.5 dB
dashed: harmonics
dotted: (k+½)F0
0
−10
−20
−30
−40
−50
dB re max
½
1
2
3
4
frequency / local F0